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parameter : N = 20, a = 2, b = 0, k = 3, theta θ = 60°, alpha α = 0,001

5. dik :

$a = 2$

$b = 0$

$k = 3$

$$L_1 = \lim_{x \to k} \left[\frac{x^2 - k^2}{x - k} \right] = \lim_{x \to 3} \left[\frac{x^2 - 3^2}{x - 3} \right]$$

$$L_2 = \lim_{x \to 0} \left[\frac{\sqrt{x + (k+1)^2} - (k+1)}{x} \right] = \lim_{x \to 0} \left[\frac{\sqrt{x + (3+1)^2} - (3+1)}{x} \right] = \lim_{x \to 0} \left[\frac{\sqrt{x+16} - 4}{x} \right]$$

$$L_3 = \lim_{x \to \infty} \left[\frac{(a+2)x^3 + bx + 7}{x^3 - x + k} \right] = \lim_{x \to \infty} \left[\frac{(2+2)x^3 + 0x + 7}{x^3 - x + 3} \right] = \lim_{x \to \infty} \left[\frac{4x^3 + 7}{x^3 - x + 3} \right]$$

(a) hitung manual

→ perhitungan L_1 (metode pemfaktoran)

$$L_1 = \lim_{x \to 3} \frac{x^2 - 9}{x - 3}$$

Faktorkan pembilang dengan selisih kuadrat

$$L_1 = \lim_{x \to 3} \frac{(x-3)(x+3)}{x-3}$$

eliminasi faktor $x - 3$

$$L_1 = \lim_{x \to 3} (x + 3)$$

Substitusi $x = 3$

$$L_1 = 3 + 3 = 6$$

→ Perhitungan L_2 (metode rasionalisasi)

$$L_2 = \lim_{x \to 0} \frac{\sqrt{x+16} - 4}{x}$$

kalikan dengan konjugat

$$L_2 = \lim_{x \to 0} \frac{\sqrt{x+16} - 4}{x} \cdot \frac{\sqrt{x+16} + 4}{\sqrt{x+16} + 4}$$

$$L_2 = \lim_{x \to 0} \frac{(x+16) - 16}{x(\sqrt{x+16} + 4)}$$

$$L_2 = \lim_{x \to 0} \frac{\cancel{x}}{\cancel{x}(\sqrt{x+16} + 4)}$$

eliminasi x dan substisi $x = 0$

$$L_2 = \frac{1}{\sqrt{0+16} + 4} = \frac{1}{4+4} = \frac{1}{8}$$

PAPERLINE

Dipindai dengan CamScanner